

FLUIDMIND SCIENTIFIC REFERENCE COMPENDIUM

OFFICIAL TEXTBOOK & LECTURE REPOSITORY • VOL. 2026

APPLIED ENGINEERING THERMODYNAMICS

THERMODYNAMICS: AN ENGINEERING APPROACH

Author / Compiler: Yunus A. Cengel & Michael A. Boles

Curated by: Mohammadamin Sharif (IUST Mechanical Engineering Department)

VOLUME OVERVIEW & ABSTRACT:

Premier reference on the First and Second Laws of Thermodynamics, closed and open control volume energy analysis, entropy generation, exergy analysis, Brayton power cycles, Rankine vapor cycles, and refrigeration systems.

SYLLABUS & CORE ENGINEERING TOPICS COVERED:

- Rigorous mathematical derivations & fundamental physical laws
- Complete worked analytical solutions with free-body & control volume diagrams
- Numerical verification & computational correlation (CFD / FEA / MATLAB)
- Practical engineering applications, safety factors, and standard design codes
- Comprehensive reference tables, property charts & dimensional constants
- Formatted for academic study, research references, and professional exam preparation

FUNDAMENTAL EQUATIONS, WORKED METHODOLOGY & NOTES

1. Governing Equations & Control Volume Formulations

Conservation of mass, linear and angular momentum, and first law energy conservation applied across differential fluid elements and me

2. Boundary Conditions & Interface Dynamics

Evaluation of kinematic velocity continuity, no-slip wall conditions, thermal heat flux balances, and dynamic shear equilibrium across conti

3. Dimensionless Numbers & Similitude Criteria

Reynolds (Re), Mach (Ma), Prandtl (Pr), Nusselt (Nu), and Euler (Eu) scaling analysis ensuring experimental and numerical dynamic simi

4. Computational Verification & Solution Strategy

Iterative finite volume / finite element procedures, discretization error bounds, convergence criteria, and cross-comparison against empiric